



EUNOIA JUNIOR COLLEGE
JC2 Preliminary Examination 2019
General Certificate of Education Advanced Level
Higher 3

MATHEMATICS

9820/01

24 September 2019

Paper 1 [100 marks]

3 hours

Additional Materials: Answer Booklet
List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **7** printed pages.

- 1 (i) Find the general solution of the differential equation

$$\frac{dx}{dt} = \frac{(c+t)x}{1-t^2}, \quad (\text{A})$$

where c is a constant. [5]

- (ii) Make the substitution $y = tx$ in the differential equation

$$\frac{dy}{dx} = \frac{ax+by}{bx+ay}, \quad (\text{B})$$

where a and b are constants and $a \neq 0$, and show that the resulting equation is equivalent to equation (A). [3]

- (iii) Hence show that equation (B) has solutions of the form

$$|x-y|^{b+a} = D|x+y|^{b-a},$$

where D is an arbitrary constant. [3]

- 2 (i) Show that $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$ for any function f which is continuous on the domain $[a, b]$. [1]

- (ii) Using the result in (i), show that $\int_0^{\frac{\pi}{4}} \ln(1+\tan \theta) d\theta = \int_0^{\frac{\pi}{4}} \ln\left(\frac{2}{1+\tan \theta}\right) d\theta$. [2]

- (iii) Hence, by using an appropriate substitution, find the exact value of $\int_0^1 \frac{\ln(1+x)}{1+x^2} dx$. [3]

- (iv) Use the result in (i) to also find the exact value of $\int_0^{\frac{\pi}{4}} \frac{\cos x}{\cos(x-\frac{\pi}{4})} dx$. [3]

- 3 (a) Let x, y, z be real numbers such that $0 < x < y < z$. Show that x^2, y^2, z^2 are three consecutive terms of an arithmetic sequence if and only if $\frac{1}{y+z}, \frac{1}{x+z}, \frac{1}{x+y}$ are three consecutive terms of another arithmetic sequence. [3]

- (b) An infinite sequence of integers, $a_n, n \in \mathbb{Z}^+$, is defined by $a_1 = a_2 = 1$ and

$$a_{n+1} = \frac{a_n^2 + 2}{a_{n-1}}, \quad n \geq 2.$$

- (i) Show that $\frac{a_{n+2} + a_n}{a_{n+1}} = \frac{a_{n+1} + a_{n-1}}{a_n}$ for $n \geq 2$. [2]

- (ii) Hence or otherwise, show that $a_{n+1} = 4a_n - a_{n-1}$ and explain why a_n is an odd integer for all $n \in \mathbb{Z}^+$. [2]

- (c) Prove that there is no arithmetic progression which has $1, \sqrt{2}, \sqrt{3}$ among its terms. [5]

- 4 Let n be a positive integer greater than 2 and $x_0, x_1, x_2, \dots, x_n$ be real numbers with $x_0 = 1$.

- (i) By using the Cauchy-Schwarz Inequality or otherwise, show that

$$\sum_{i=1}^n (x_{i-1} - x_i)^2 + x_n^2 \geq \frac{1}{n+1}. \quad [3]$$

- (ii) Given that $\sum_{i=1}^n (x_{i-1} - x_i)^2 + x_n^2 = \frac{1}{n+1}$, find x_k for $0 \leq k \leq n, k \in \mathbb{Z}$, leaving your answer in terms of n and k . [3]

- 5 A factory supervisor intends to distribute work to his three subordinates. He has 24 indistinguishable cards each representing one unit of work, and he intends to distribute them into three different colored boxes, red, green and blue, representing each of his subordinates.

(i) Find the number of ways he can distribute the cards into the boxes without restriction. [1]

The supervisor wishes to split the work relatively evenly amongst his three subordinates so that each subordinate has at most 10 units of work.

(ii) Find the number of ways the supervisor can distribute the workload. [3]

The red box represents the workload of Sue, who is pregnant, and the supervisor will like to reduce her workload such that she will only have a workload at most half of any of her other two colleagues. For example, if Sue is given 3 units of work, then her colleagues must have at least 6 units of work each.

(iii) Find the number of ways the supervisor can distribute the workload. [3]

Peter is given r units of work. He places the r indistinguishable cards into n indistinguishable boxes for him to randomly choose one box for each of the subsequent days.

Let the number of ways for him to do so be denoted by $P(r, n)$.

(iv) State the recurrence relation between $P(r, n)$, $P(r-1, n-1)$ and $P(r-n, n)$. [1]

(v) Using the result in (iv) or otherwise, show that $P(r, n) = \sum_{k=1}^n P(r-n, k)$ for $r > n \geq 1$. [2]

(vi) Using (iv) or (v), find the value of $P(10, 3)$. [2]

- 6** You have an unlimited supply of 1×1 , 1×2 and 2×2 tiles. Tiles of the same size are indistinguishable.

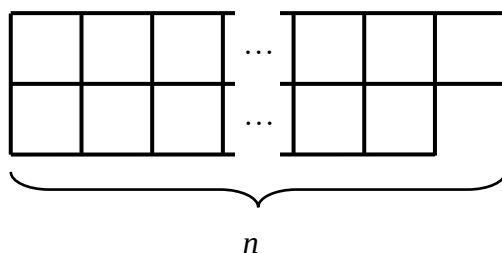
(i) Let T_n is the number of ways of tiling a $1 \times n$ path.

(a) State the value of T_1 and T_2 . [1]

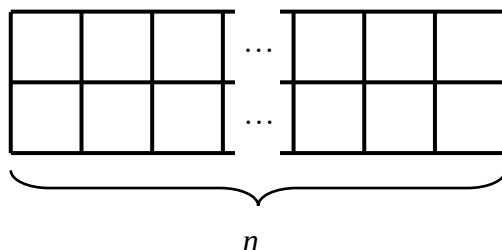
(b) Write down an appropriate recurrence relation between T_{n+2} , T_{n+1} and T_n . [1]

Consider the tilings of a $2 \times n$ path. (The 1×2 tiles can be rotated in the tilings.)

Let P_n be the number of tilings of



Let Q_n be the number of tilings of



(ii) Show that $P_{n+1} = P_n + Q_n$ for $n \geq 1$. Explain your reasoning clearly. [2]

(iii) Show that $Q_{n+1} = 2P_{n+1} + 2Q_{n-1}$ for $n \geq 2$. Explain your reasoning clearly. [4]

(iv) Use **(ii)** and **(iii)** to show that $P_{n+2} + 2P_{n-1} = 3P_{n+1} + 2P_n$ for $n \geq 2$. [2]

It is given that the solution to the above recurrence relation is

$$P_n = -\frac{(-1)^n}{7} + \frac{1+2\sqrt{2}}{14}(2+\sqrt{2})^n + \frac{1-2\sqrt{2}}{14}(2-\sqrt{2})^n.$$

(v) Find the number of distinct ways of tiling a $2 \times n$ path. [2]

- 7 (a) Show that if $7|(a^2 + b^2)$ then $7|a$ and $7|b$ where a and b are positive integers. [3]
- (b) Given that n is an integer greater than 1, by considering the cases when n is even and n is odd, show that $n^4 + 4^n$ cannot be a prime number. [4]
- (c) Show that for every positive integer n , $16^n + 10n - 1$ is divisible by 25. [5]

8 Let
$$L_1(x) = \frac{(x - x_2)(x - x_3)}{(x_1 - x_2)(x_1 - x_3)}.$$

- (i) State the values of $L_1(x_1)$, $L_1(x_2)$ and $L_1(x_3)$. [2]
- (ii) Hence or otherwise, obtain expressions of quadratic polynomials L_2 and L_3 such that $L_i(x_i) = 1$, and $L_i(x_j) = 0$, $i \neq j$, $i = 2, 3$, and $j = 1, 2, 3$. [2]

It is known that if p is a polynomial of degree m , and $p(x_i) = 0$, for $i = 1, 2, \dots, m+1$, where $x_1 < x_2 < x_3 < \dots < x_n < x_{m+1}$, then $p(x) = 0$, for all $x \in \mathbb{R}$.

- (iii) Suppose there are 3 distinct values x_1, x_2, x_3 and any real values y_1, y_2, y_3 . Using the above result, explain why there is at most one quadratic polynomial P such that $P(x_i) = y_i$ for $i = 1, 2, 3$. [3]
- (iv) Obtain the quadratic polynomial P as a combination of y_i and L_i such that $P(x_i) = y_i$ for $i = 1, 2, 3$, and justify your answer. [2]
- (v) Let $x_i = i$ for $i = 1, 2, 3$, and $\{y_1, y_2, y_3\} = \{1, 2, 3\}$. Find all possible quadratic polynomials Q satisfying $Q(x_i) = y_i$, $y_i \neq i$, for $i = 1, 2, 3$, with appropriate justification to be provided. [You need not simplify your answer] [4]

- 9 Let $S(n)$ denote the sum of all divisors of the number n . For example,
 $S(6) = 1 + 2 + 3 + 6 = 12$.

(i) Find

(a) $S(p)$ in terms of p , [1]

(b) $S(pq)$ in terms of p and q , [1]

(c) $S(p^m q^n)$ in terms of p, q, m and n , [2]

where p, q are distinct prime numbers and m, n are positive integers.

Aliquot divisors of a number are divisors of the number excluding the number itself (i.e. the sum of all proper divisors of the number). For example, the aliquot divisors of 6 are 1, 2 and 3 respectively.

Two numbers are called amicable if each equals the sum of the aliquot divisors of the other, and the two numbers form an amicable pair.

(ii) Verify that 220 and 284 is an amicable pair. [2]

Assume that two amicable numbers, M and N , has the form $M = apq$ and $N = ar$, where p, q, r are distinct prime numbers, a (not necessarily prime) is the common divisor of M and N , and a is co-prime to p, q and r .

(iii) Show that $a(pq + r) = (p + 1)(q + 1)S(a)$. [2]

(iv) Express $r + 1$ in terms of p and q only, and use (iii) to show that
 $(p + 1)(q + 1)[2a - S(a)] = a(p + q + 2)$. [3]

The equation in (iv) can be further manipulated to the following equality:

$$\left(p + 1 - \frac{a}{2a - S(a)}\right) \left(q + 1 - \frac{a}{2a - S(a)}\right) = \left(\frac{a}{2a - S(a)}\right)^2$$

(v) Find the only possible solution for p and q for $a = 4$, and hence find M and N . [2]

End of Paper